Lecture 08: Generative Models and Clustering

Machine Learning Tutorial

School of Engineering, Westlake University

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Outline

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1.1 Generative Classification and GDA

Known labels identify the groups used for fitting.

Observed classes



Gaussian input model

$$p(x \mid y = c) = \mathcal{N}(x \mid \mu_c, \Sigma_c).$$

Count the class, then average its inputs

$$\hat{\pi}_c = \frac{N_c}{N}, \qquad \hat{\mu}_c = \frac{1}{N_c} \sum_{n:y_n=c} x_n.$$

N_c: class-c inputs. N: all inputs.

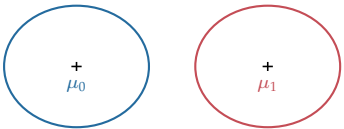
$$\underbrace{w_c(x) = \hat{\pi}_c \mathcal{N}(x \mid \hat{\mu}_c, \hat{\Sigma}_c)}_{\text{prior times class-conditional density}}, \qquad p(y = c \mid x) = \frac{w_c(x)}{\sum_j w_j(x)}.$$

Fit class distributions first. Use Bayes' rule to classify a new input.

1.2 Covariance Estimation: LDA and QDA

Subtract each input's own class mean: $r_n = x_n - \hat{\mu}_{y_n}$.

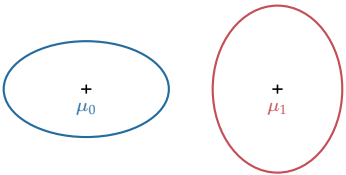
LDA: one shared covariance



$$\hat{\Sigma} = \frac{1}{N} \sum_{n=1}^{N} r_n r_n^\top .$$

Pool the within-class residuals.

QDA: one covariance per class



$$\hat{\Sigma}_c = \frac{1}{N_c} \sum_{n:y_n=c} r_n r_n^\top .$$

Estimate spread separately within each class.

Colored ellipses are equal-density contours. Covariance controls their spread and orientation.

1.3 Decision Boundaries: Equal Class Posteriors

For two classes, the boundary contains inputs where the posteriors tie:

$$p(y = 1 \mid x) = p(y = 0 \mid x) = \tfrac{1}{2} \quad \Longleftrightarrow \quad \pi_1 p(x \mid y = 1) = \pi_0 p(x \mid y = 0).$$

Use a log score, dropping the same additive constant for both classes:

$$g_c(x) = \log \pi_c - \frac{1}{2} \log |\Sigma_c| - \frac{1}{2}(x - \mu_c)^\top \Sigma_c^{-1}(x - \mu_c).$$

Expand the Gaussian quadratic form:

$$g_c(x) = \underbrace{-\frac{1}{2} x^\top \Sigma_c^{-1} x}_{\text{quadratic in } x} + \underbrace{\mu_c^\top \Sigma_c^{-1} x}_{\text{linear in } x} + \underbrace{a_c}_{\text{constant in } x},$$

$$a_c = \log \pi_c - \frac{1}{2} \log |\Sigma_c| - \frac{1}{2} \mu_c^\top \Sigma_c^{-1} \mu_c.$$

The decision boundary solves $g_1(x) - g_0(x) = 0$.

1.4 Linear and Quadratic Decision Boundaries

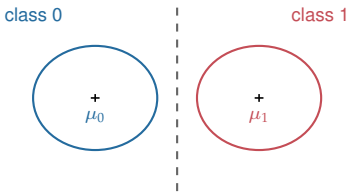
$$g_1(x) - g_0(x) = x^\top \underbrace{\frac{1}{2}(\Sigma_0^{-1} - \Sigma_1^{-1})}_{A} x + w^\top x + b.$$

$$w = \Sigma_1^{-1}\mu_1 - \Sigma_0^{-1}\mu_0, \qquad b = a_1 - a_0.$$

LDA: $\Sigma_0 = \Sigma_1$

The quadratic terms cancel.

$$w^\top x + b = 0$$

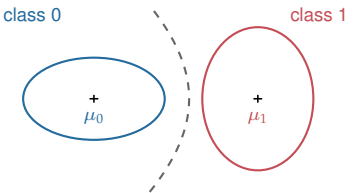


A line in 2D, a hyperplane in higher dimensions.

QDA: class-specific Σ_c

Quadratic terms generally remain.

$$x^\top A x + w^\top x + b = 0$$



A degree-two equation, generally a curved boundary.

Colored ellipses: within-class density contours. Gray dashed line: equal class posteriors.

1.5 Bernoulli Naive Bayes: Definition

A generative classifier for binary inputs $x \in \{0,1\}^D$ and class $y \in \{1,\dots,C\}$.

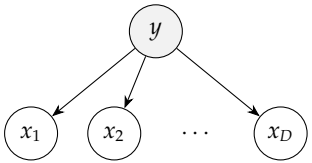
Class prior $p(y=c) = \pi_c,\ \pi_c \geq 0,\ \sum_c \pi_c = 1.$

Bernoulli: one probability per binary feature

$$x_j \mid y = c \sim \text{Bernoulli}(q_{jc}), \quad \textit{i.e.}, q_{jc} = p(x_j = 1 \mid y = c) \in [0,1].$$

Naive: features are independent given y

$$p(x \mid y = c) = \prod_{j=1}^{D} p(x_j \mid y = c).$$



Presence or absence
Dictionary: [offer, meeting]
"offer offer" gives $x = (1,0)$.

Each q_{jc} describes one word's presence. The q_{jc} values need not sum to one over words.

$$\underbrace{p(x, y=c)}_{\text{joint model}} = p(y=c) \cdot p(x \mid y=c) = \underbrace{\pi_c}_{\text{class prior}} \prod_{j=1}^{D} \underbrace{q_{jc}^{x_j}(1-q_{jc})^{1-x_j}}_{\text{Bernoulli probability of feature } j} .$$

Bayes' rule turns this joint model into the class posterior $p(y=c \mid x)$.

1.6 Bernoulli Naive Bayes: Training and Prediction

Training: count within each observed class

$$\hat{\pi}_c = \frac{N_c}{N}, \qquad \hat{q}_{jc}^{\text{MLE}} = \frac{N_{jc}}{N_c}, \qquad \underbrace{\hat{q}_{jc}^{\text{Laplace}} = \frac{N_{jc}+1}{N_c+2}}_{\text{one present and one absent pseudo-count}} .$$

N_c: class-c inputs. $N_{jc} = \sum_{n:y_n=c} x_{nj}$: class-$c$ inputs containing word j.

Example fitted probabilities

Class	offer	meeting
Spam	.8	.2
Non-spam	.1	.7

Email: "offer" $x = (1, 0)$

$\hat{\pi}_{\text{spam}} = \hat{\pi}_{\text{nonspam}} = .5.$

$$w_{\text{spam}} = p(x, \text{spam}).5 \times .8 \times (1 - .2) = .32,$$
$$w_{\text{nonspam}} = p(x, \text{nonspam}) = .5 \times .1 \times (1 - .7) = .015.$$

$p(\text{spam} \mid x) = .32/(.32 + .015) \approx .955.$
$p(\text{nonspam} \mid x) = .015/(.32 + .015) \approx .045.$

$$p(\text{spam} \mid x) > p(\text{nonspam} \mid x) \Rightarrow \text{spam}.$$

2.1 K-means: Objective and Pseudocode

$$\min_{z,\mu} J(z,\mu) = \sum_{n=1}^{N} \|x_n - \mu_{z_n}\|^2, \qquad z_n \in \{1,\dots,K\}.$$

Require: Inputs $x_1,\dots,x_N \in \mathbb{R}^D$ and $1 \le K \le N$.
1: Randomly sample K inputs without replacement as centers $\mu_1,\dots,\mu_K$.
2: **repeat**
3: Save the current centers: $\mu^{\text{old}} \leftarrow \mu$.
4: **for** each input x_n **do**
5: **Assign:** $z_n \leftarrow \arg\min_{1\le k\le K} \|x_n - \mu_k^{\text{old}}\|^2$.
6: **end for**
7: **for** each nonempty cluster k **do**
8: **Update:** $\mu_k \leftarrow$ mean of the inputs assigned to cluster k.
9: **end for**
10: **until** all centers stay unchanged: $\mu = \mu^{\text{old}}$
11: **return** assignments z and centers μ.

Empty clusters keep their previous centers. Break distance ties by choosing the smallest cluster index.

3.1 Gaussian Mixture Models

GDA observes the group labels. A GMM treats the labels as hidden variables.

Assume K Gaussian components.

1. Choose one hidden component

$$\Pr(z = k) = \pi_k, \qquad \sum_{k=1}^{K} \pi_k = 1.$$

2. Draw an input from its Gaussian

$$x \mid z = k \sim \mathcal{N}(\mu_k, \Sigma_k).$$

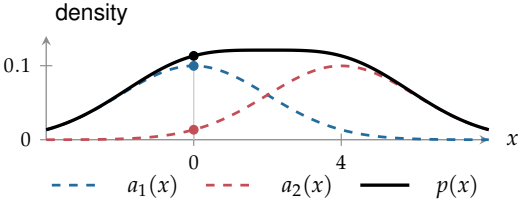
π_k chance of choosing component k

μ_k component center

Σ_k spread and orientation

Example: assume $K = 2$

$$\pi_1 = \pi_2 = .5, \quad \mu = (0, 4), \quad \sigma_1^2 = \sigma_2^2 = 4.$$



The total curve is a density with area 1.

Joint density describes one source. Marginal density adds all sources.

$$\underbrace{a_k(x) = p(x, z = k \mid \theta)}_{\text{joint density for source } k} = \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k), \qquad \underbrace{p(x \mid \theta) = \sum_{k=1}^{K} a_k(x)}_{\text{mixture density}}.$$

Parameters: $\theta = \{(\pi_k, \mu_k, \Sigma_k)\}_{k=1}^{K}$, with $\pi_k > 0$ and $\Sigma_k \succ 0$.

3.2 Learning a Gaussian Mixture

Learning: given only $X = (x_1, \ldots, x_N)$, estimate $\theta = \{(\pi_k, \mu_k, \Sigma_k)\}_{k=1}^{K}$.

Independent inputs: multiply their mixture densities

$$L(\theta; X) = \prod_{n=1}^{N} p(x_n \mid \theta).$$

Maximum likelihood: maximize the log-likelihood over θ

$$\ell(\theta; X) = \sum_{n=1}^{N} \underbrace{\log \left[\sum_{k=1}^{K} \pi_k \mathcal{N}(x_n \mid \mu_k, \Sigma_k) \right]}_{\text{one input: log of the sum over its possible sources}} .$$

If the labels were known

One selected term per input:

$$\log a_{z_n}(x_n).$$

Fit each labeled group separately.
Estimate its proportion, center and spread.

With hidden labels

A sum over all possible sources:

$$\log \left[\sum_{k=1}^{K} a_k(x_n) \right].$$

Each input involves all components.
Estimate their parameters jointly.

The log contains a sum, so the Gaussian fits are coupled.

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